

# The Interview-Compatible Recommender: A Measuring Instrument for Cold-Start Conversational Recommendation

Thesis Chapter A (v2)

July 2026

## Abstract

Before one can study *how* to interview a cold-start user—which questions to ask, in what order—one needs a recommender that can *be* interviewed: a model that folds an arbitrary-length, mixed-channel set of answers into its belief and re-ranks a large catalogue, without per-user retraining and without trading away full-profile accuracy. We argue that this recommender is a *measuring instrument* for the entire elicitation-policy literature, and that its requirements are more demanding than any single published system meets. We state five requirements—**R1** state-of-the-art full-profile accuracy, **R2** arbitrary-length cold-start fold-in, **R3** channel-agnostic input (items, out-of-catalogue concepts, entities, and continuous directions through one interface), **R4** an uncertainty-native belief, and **R5** monotonicity (a truthful answer never lowers ranking quality)—and an acceptance battery (per-build gates **G0–G9** plus one-time certifications **C1–C3**) that operationalises them. A survey of six model families shows the literature is *bifurcated*: the models that hit the full-profile bar (shallow linear/autoencoder recommenders such as EASE [33] and RecVAE [31]) are item-only point estimators, while the models with an uncertainty-native, multi-channel belief (soft-attribute Gaussian elicitation [4], unified item+attribute bandits [17], conjugate critique folds [36]) do not report full-profile accuracy at the R1 bar. We are aware of no published system that occupies the conjunction of all five. The instrument we propose to close this gap is a *frozen* certified shallow-SOTA tower with an analytic conjugate-Gaussian belief layer over its latent, into which every channel folds as a linear observation; three of its input legs—signed/graded values, out-of-catalogue concepts, and continuous direction tokens—are architecturally inexpressible in the item-indicator basis of the R1-bar models. This chapter contributes the requirement framework, the taxonomy and gap analysis, the instrument’s design, and the experimental protocol (a metric-certification bridge to the published ML-20M numbers, then G0/G1/G2 on the ML-25M harness). Our in-house RecVAE realisation sets the bar at full/tail NDCG@10 0.3540/0.2497 on the canonical ML-25M protocol, and the graded-native interview tower now *meets* it: a single certified training run reaches 0.3482/0.2462 on the held-out test cohort—a paired-bootstrap tie with the bar—while reproducing the frozen decoder’s ranking exactly at the empty interview, so G0 (full strength preserved) is passed rather than assumed. The interview-side results (the graded-input curve, the concept channel) are measured on the same architecture but on a pre-certification snapshot and an earlier fold recipe; they are labelled as such throughout and their re-runs on the certified tower are queued.

## Contents

### 1 Introduction: the recommender as a measuring instrument

2

|     |                                                                                   |    |
|-----|-----------------------------------------------------------------------------------|----|
| 2   | Requirements and gates                                                            | 4  |
| 3   | Related work: a six-family taxonomy                                               | 6  |
| 4   | Gap analysis                                                                      | 8  |
| 5   | The instrument                                                                    | 9  |
| 5.1 | Frozen shallow-SOTA tower . . . . .                                               | 9  |
| 5.2 | Analytic conjugate-Gaussian belief layer . . . . .                                | 9  |
| 5.3 | Channel-agnostic tokens as linear observations . . . . .                          | 10 |
| 6   | The concept channel: derivation, lineage, and failure analysis                    | 10 |
| 6.1 | A signed, exposure-corrected affinity estimator . . . . .                         | 11 |
| 6.2 | The fold operator: whitened directions and a gated fold-to-point . . . . .        | 12 |
| 6.3 | Failure analysis: a positive-only clip and the deployment-currency gate . . . . . | 13 |
| 6.4 | Results: the signed retrain, and where the concept channel wins . . . . .         | 14 |
| 7   | Experimental design                                                               | 16 |
| 7.1 | Baseline bank . . . . .                                                           | 16 |
| 7.2 | Evaluation protocol . . . . .                                                     | 17 |
| 8   | Limitations and threats to validity                                               | 19 |

# 1 Introduction: the recommender as a measuring instrument

A *cold-start* conversational recommender must elicit taste from a short interview and then rank a large catalogue. Two questions are entangled: *what to ask* (the elicitation policy) and *how to turn answers into recommendations* (the recommender). The elicitation-policy literature is large and active, but almost every comparison in it is confounded: a policy’s apparent value depends on the recommender it drives, and reported gains routinely conflate a better *question-selection strategy* with a better *answer-ingestion model*. Before we can measure elicitation strategies in isolation, we need to fix the recommender—and not just any recommender, but one strong and flexible enough that a good interview is not wasted on it. We call such a recommender an *interview-compatible instrument*.

**Five requirements.** An instrument fit for measuring conversational elicitation must simultaneously satisfy:

- R1 — full-profile SOTA.** On a full interaction history it must rank at the level of the strongest shallow collaborative-filtering baselines (EASE [33], RecVAE [31]), so that elicitation is measured against a recommender nobody can dismiss as weak [10].
- R2 — any-length cold-start fold-in.** It must ingest an evidence set of *any* cardinality, from 0 (fully cold) upward, without per-user retraining.
- R3 — channel-agnostic input.** Items, out-of-catalogue concepts, entities, and free text (and, for the continuous policy, direction tokens) must all fold through *one* interface as arbitrary embeddings, not a fixed categorical schema.
- R4 — uncertainty-native belief.** It must carry an explicit covariance over the user belief, not just a point estimate, so that a policy can act on *what it does not yet know*.

**R5 — monotonicity.** A truthful answer must never *lower* ranking quality; adding evidence should, in expectation, only help.

**The bifurcation observation.** The central empirical fact this chapter documents is that R1 and {R3, R4, R5} are *anti-correlated across the published literature*. The models that hit the full-profile bar—closed-form linear autoencoders (EASE [33], EDLAE [34]), amortised VAEs (Mult-VAE [19], RecVAE [31]), and training-free graph filters (GF-CF [30], BSPM [7], Turbo-CF [26])—are *item-only point estimators* on a one-hot indicator basis. The models with an uncertainty-native, multi-channel belief—soft-attribute Gaussian elicitation [4], item+attribute Thompson bandits [17], conjugate critique folds over a frozen factorisation [36], and per-item Bayesian NL elicitation [3]—do *not* report full-profile accuracy at the R1 bar. We call these the *accuracy corner* and the *belief corner*. No published system we are aware of occupies both (§4).

**Contributions.** This chapter is the instrument paper of the thesis. It does not claim a new elicitation policy (that is the companion method chapter); it claims the *recommender that makes policy measurement possible*. Concretely:

1. **The conjunction as the contribution.** We frame  $R1 \wedge R2 \wedge R3 \wedge R4 \wedge R5$  as a single requirement no published system meets, and give a six-family taxonomy and per-near-miss gap analysis (§3, §4) that locates exactly which requirement each candidate fails. We say we are *aware of none*, never that there *is* none.
2. **Three architecturally-inexpressible legs.** We isolate the three input channels that are impossible (not merely untuned) in the item-indicator basis of every R1-bar model: (i) signed/graded per-entity values, (ii) out-of-catalogue concepts, and (iii) continuous direction tokens. These are the narrow, defensible novelty legs of the channel-agnostic interface (§4, §5).
3. **The instrument.** A design for a *frozen* certified shallow-SOTA tower with an analytic conjugate-Gaussian precision-accumulator belief layer over its latent, into which every channel folds as a linear observation (§5). The frozen-tower+exact-conjugacy shape is the one mechanism with a principled monotonicity story [36], and it is what separates us from the co-trained, approximate-posterior near-miss of Büyük et al. [4].
4. **An acceptance battery as a reusable certificate.** We define three headline outcome gates—full-strength preserved (G0), posterior strictly shrinks per question (G1), and cold-start per-question NDCG rises on *full and tail* versus random (G2)—and, because these outcome gates admit several false positives (a sign-blind fold, a circular answer model, a decorative covariance), extend them into a full battery of per-build gates G0–G9 plus one-time certifications C1–C3 that add the semantic, artifact-control, and answer-model-transfer checks. We propose the battery as a portable certificate any interview-compatible recommender should pass (§2, §7).

We are careful throughout to distinguish what is *new* (the conjunction; the frozen-certified-tower wedge; the acceptance battery on a SOTA tower) from what is *pre-empted* and must be cited (attributes-as-directions plus a Bayesian fold [4]; unified item+attribute belief [17]; conjugate folds over a frozen factorisation [36]; permutation-invariant set-encoders with information-gain acquisition [24]; value-carrying set-input cold-start recommenders [20]; the non-degradation-of-information principle [12, 5]). “Uncertainty-native recommender” alone is a crowded 2024–25 area [6] and is *not* claimed as novel.

## 2 Requirements and gates

We now state the requirements and gates formally. Let the catalogue have  $n$  items and let the frozen tower expose a  $d$ -dimensional latent in which each item  $i$  has a factor row  $\mathbf{Q}_i \in \mathbb{R}^d$ . An *evidence set* after  $t$  questions is a set of observations  $\mathcal{E}_t = \{(\phi_j, y_j)\}_{j=1}^t$ , where  $\phi_j \in \mathbb{R}^d$  is the latent direction of the queried entity (an item factor, a concept centroid, an entity embedding, or a continuous direction token) and  $y_j \in \mathbb{R}$  is the (possibly graded, possibly signed) answer. The instrument maintains a belief  $b(\mathcal{E}_t) = (\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$  over the user latent  $\mathbf{u} \in \mathbb{R}^d$  and scores item  $i$  by a fixed read-out  $s_i(\boldsymbol{\mu}_t)$  of the base tower.

### Requirements (formal).

- R1.** With  $\mathcal{E}$  equal to a user’s full history,  $\text{NDCG@10}(s(\boldsymbol{\mu}(\mathcal{E})))$  ties the strongest shallow baseline on a shared harness (Def. 1).
- R2.**  $b(\mathcal{E}_t)$  is defined and computable for every  $t \geq 0$  with no gradient step at inference;  $t=0$  yields the prior belief (a non-personalised ranker).
- R3.** The update from  $\mathcal{E}_t$  to  $\mathcal{E}_{t+1}$  depends on the new observation only through the pair  $(\phi_{t+1}, y_{t+1}) \in \mathbb{R}^d \times \mathbb{R}$ ; the operator is identical for every channel, so any entity with an embedding is a legal observation—including entities absent from the catalogue.
- R4.**  $\boldsymbol{\Sigma}_t$  is an explicit, queryable covariance; a policy may read  $\boldsymbol{\Sigma}_t$  to choose the next question.
- R5.** For a truthful answer  $y_{t+1}$ ,  $\mathbb{E}[\text{NDCG@10}(s(\boldsymbol{\mu}_{t+1}))] \geq \text{NDCG@10}(s(\boldsymbol{\mu}_t))$ . We treat R5 as an *empirical* property to be demonstrated per step, not a theorem for a learned ranker (§8).

**Definition 1** (Ties SOTA). *A recommender ties a baseline on a metric if their scores are within the paired per-user bootstrap confidence interval on one shared harness (same users, items, split, metric depth). A number computed on a different metric/split (e.g. NDCG@10 on ML-25M vs. NDCG@100 on the ML-20M strong-generalisation split) is not a tie until bridged (§7.2).*

**Gates (acceptance battery).** The three headline gates below (G0/G1/G2) are *outcome* gates: they certify that the full-profile strength is preserved, that the posterior is usable, and that the cold-start curve rises. On their own they are insufficient (see below), so the instrument is accepted only against the full battery—per-build gates G0–G9 (every build, all cheap) and three one-time certifications C1–C3. Table 1 is the compact reference; the definitions follow.

*Per-build gates.*

- G0 — full strength & identity.** Evaluating on the belief mean with the full profile reproduces the frozen tower’s full-profile NDCG@10 (full *and* tail)—a paired-bootstrap tie, or bit-identity under the frozen-fold design—so the belief layer adds no full-profile regression (certifies R1 survives).
- G1 — usable posterior.** The posterior covariance shrinks in expectation over confident answers, with *refusals explicitly exempt* (a correct model does not shrink on a refusal); the shrinkage is directional (at least twice as large along the queried  $\phi_{t+1}$  as elsewhere) and the belief scale is calibrated against held-out NLL (certifies R4).
- G2 — cold-start curve rises on full and tail.** Starting cold, per-question full and tail NDCG@10 beats a random-question control judged by the *curve/endpoint at budget* (an AUC criterion, *not* every-step significance), and must strictly *increase* from the first to the last turn (not

merely fail to drop); plus an out-of-envelope canary that a single true answer from cold helps on *every* channel. This is the operational form of R2∧R5.

- G3 — sign & intensity fidelity.** Negating every answer’s sign lowers NDCG (CI excluding zero, per channel); neutralising values to “indifferent” while freezing membership costs NDCG; and quality is monotone in stated intensity, so a binarising ablation must lose.
- G4 — confidence & refusal.** The fitted per-answer precisions order  $\alpha(\text{refuse}) < \alpha(\text{vague}) < \alpha(\text{know-well})$ , collapsing the confidence levels loses NDCG, and a refusal burns the turn while moving the belief by approximately nothing.
- G5 — concept/channel specificity.** Folding a concept ranks its member items above matched non-members (member-lift AUC > 0.8) *and* beats that concept’s own popularity projection (top principal component partialled out), with a dedicated channel at least matching a trained member-bag injection.
- G6 — existential controls.** Wrong-user, shuffled, and placebo-constant answers through the identical pipeline gain approximately nothing relative to the true-answer arm, and duplicating an answer changes nothing (the belief responds to information, not cardinality).
- G8 —  $\Sigma$  load-bearing & set coherence.** Replacing  $\Sigma$  with an isotropic  $c\mathbf{I}$  *degrades* the acceptance-deciding number (else the R4 claim is withdrawn), and the belief and ranking are invariant to answer order and to conflicting re-asks.
- G9 — answer-ingestion isolation.** On a *fixed*, user-independent question set the curve still rises with  $t$  from answer content alone, separating instrument *ingestion* from question-selection *policy*.

*One-time certifications* (per instrument, not per build):

- C1 — published-number bridge.** The implementations snap to the published full-profile numbers on the canonical split (*met, with one disqualification*: EASE and RecVAE reproduce to the third decimal on the ML-20M Liang split and Mult-VAE near-snaps at  $-0.004$ , while Mult-DAE fails at  $-0.022$  and is dropped; §7.2).
- C2 — answer-model transfer (non-circularity).** The G2 gain survives under a *held-out behavioural* answer model plus injected label noise, at  $\geq$  a stated fraction of the self-model gain with a CI excluding zero, and a firewall assert guarantees that recommender-geometry answers appear nowhere in training or evaluation. This is the load-bearing certification: without it a G0–G2 pass is consistent with an instrument inverting its own simulator.
- C3 — human round-trip.** At least one study takes rendered questions  $\rightarrow$  graded human answers  $\rightarrow$  downstream NDCG. Deferred; carried as a project-wide external-validity obligation (§8).

**Why the outcome gates alone are insufficient.** G0/G1/G2 are outcome gates, and several broken instruments pass all three. A *sign-blind* fold that updates its belief from only the *direction* of the queried entity but never the answer *value* still reproduces the frozen tower (G0), still shrinks  $\Sigma$  along  $\phi$  per question (G1), and still beats a random-question control (G2)—because beating random certifies that a good question was *selected*, not that the answer was *ingested*; only the sign-flip test (G3) and the fixed-question ingestion isolation (G9) expose it. A *circular* answer model that generates answers from the instrument’s own latent geometry inflates the G2 curve with no transferable signal; only the held-out behavioural answer model and firewall of C2 close it. And a *decorative* covariance that is maintained but never enters the acceptance-deciding number

passes G0–G2 trivially; only replacing  $\Sigma$  with an isotropic matrix and demanding a degradation (G8) shows whether the uncertainty is load-bearing. The battery’s semantic and artifact-control gates (G3–G9) and the transfer certification (C2) exist precisely to close these outcome-gate false positives.

We report *both* full and tail NDCG@10 for every gate (a project reporting rule: full-only or tail-only hides the effect, which is typically much larger on the tail than on the full catalogue). Of the battery, **G0 is passed on the certified tower** (§7, Table 5); the remaining gate results in this chapter were measured on a pre-certification snapshot of the same architecture and are labelled **[RE-RUN QUEUED: ]** where their re-run on the certified tower is queued, or **[TODO: ]** where the gate has not yet been run at all.

### 3 Related work: a six-family taxonomy

We organise interview-relevant recommenders into six families and score each against R1–R5 (Table 2). The reading is consistent across families: no family is strong on both the accuracy axis (R1) and the belief-channel axis (R3–R5). Figure 1 plots the same fact quantitatively, with measured full-profile accuracy on one axis. Table 2 is the compact conceptual overview; the chapter’s anchor is the per-system *master table* (Table 3), which classifies every reasonable candidate against R1–R5 *and*, where one exists, prints its full/tail NDCG@10 on our shared ML-25M ruler—so that the bifurcation is not merely asserted from published claims but is visible in the one column of measured numbers.

**No dash is an unexplained omission.** Every unmeasured row carries one of three dispositions. (Systems with neither a measurable number nor a distinct capability profile—e.g. SLIM, strictly dominated by its closed-form successor EASE in every published comparison—are cited in the text but given no row.) *Run planned:* the graph filters and sequence models have public code and will be measured on this ruler in the next experimental wave. *Best-effort measured / planned:* for systems whose full-profile core is runnable but whose exact recipe is unpublished or benchmark-bound, we report a best-effort replication of that core with every substitution stated. Three are now *measured*, and they land weak: a Gaussian belief over an MF latent—the shared posterior core of Bıyık (no public implementation; their embedding is co-trained, so the substitution, if anything, flatters the original) and of ConTS’s Thompson posterior—measures 0.2019/0.1200 full/tail, *below* even iALS; and the node recommender that is the Golbandi tree’s core measures 0.3065/0.1895. A trained Partial-VAE for EDDI and MF-seed reconstruction for RBMF/Functional-MF remain best-effort planned. That the belief corner’s runnable cores measure this far below the accuracy corner sharpens the bifurcation rather than softening it. *Not runnable by construction:* PEBOL maintains per-item Beta posteriors over an LLM-scored shortlist of  $\sim 100$  items and contains no full-catalogue ranker to run; GATE elicits but does not rank; zero-shot LLM ranking of an 18k catalogue is neither computationally nor methodologically meaningful at full-profile scale (and its collaborative weakness is already documented [13]); BCIE requires its authors’ knowledge graph, and a port would substitute so much that the number would no longer be theirs—its conjugate-fold core is covered by the Bıyık-style best-effort arm. ICF is the one row whose full-profile core is *already* measured: its underlying model is Bayesian probabilistic MF, so its full-profile ceiling is the measured iALS row (0.2442).

**The measured-accuracy column falls as you enter the belief corner.** Read Table 3 down its measured-accuracy column. Every *competitive* number—a full/tail NDCG@10 at or near the

frontier on our shared ML-25M ruler—sits in block (a) or (b): the item-only linear and amortised recommenders (RecVAE 0.3540, EASE 0.3476, EDLAE 0.3433). The belief corner, block (c), is either *blank* or *measurably weak*. Where its cores are runnable at all they land at the bottom of the column: the shared Gaussian-belief-over-MF core of Bıyık and ConTS at 0.2019/0.1200 (below iALS and kNN), the Golbandi node recommender at 0.3065/0.1895—and every one of these carries an ✗ on R1. The rest of the corner—EDDI, RBMF/Functional-MF, BCIE, UNICORN, PEBOL, GATE, the LLM-zero-shot CRS—has a blank accuracy column entirely. This is not an accident of our harness. Most CRS and elicitation papers never report full-catalogue top-N accuracy at all (they report elicitation RMSE, cold-start MRR/MAP on a shortlist, or critique metrics), and several *cannot produce it by construction*: a per-item Beta posterior (PEBOL) or the decision tree itself (Golbandi, whose node recommender we measure only as a best-effort core) does not induce a ranking over the whole catalogue, and an LLM CRS needs an API call per candidate to score one. Conversely, every system at the frontier is item-only and point-estimate—✗ on R3 and R4. The two blocks are disjoint: no row combines a frontier accuracy number with ✓s across R3–R5. The top-right cell of the gap map—all five requirements met *and* a competitive full-profile number—is occupied only by the target row, which is a design, not a result. That empty conjunction is the chapter’s thesis, and the master table is where it is impossible to miss.

**The accuracy corner (families ii–iii).** On the canonical ML-20M strong-generalisation split (10k validation / 10k test held-out users, NDCG@100), the full-profile frontier has been *flat since 2019–2020* and is held by shallow models, not deep nets: EASE [33] at NDCG@100 0.420, MultVAE [19] at 0.426, RecVAE [31] at 0.442 (top of this split), with EDLAE [34] in the same class; SLIM (0.401) and WMF/iALS (0.386) are the classic floors. Dacrema et al. [10] is the binding warning: most deeper “SOTA” gains vanish on this exact split. The graph-filter leaders (GF-CF [30], BSPM [7], Turbo-CF [26]) are the 2023–24 speed+accuracy front on Gowalla/Yelp/Amazon but do not report this ML-20M split, so they are admissible to our accuracy tier only if re-run on our harness. All of family (ii)/(iii) are item-only: their input is a one-hot (or set-indicator) interaction vector, which *cannot encode* a signed value, an out-of-catalogue concept, or a continuous direction—R3 is an architecture property here, not a tuning gap.

**The belief corner (family iv, and vi).** Bıyık et al. [4] maintain a multivariate-Gaussian belief over a user embedding and treat both item comparisons and “soft attributes” (e.g. “scarier”) as directions (concept-activation vectors) in the item space, selecting queries by expected value of information—the nearest thing to R2+R3+R4 together. ConTS [17] unifies items and attributes as Thompson-sampling arms with a posterior over a cold-start user. BCIE [36] folds critiques into a *frozen* factorisation by exact conjugate-Gaussian updates—the closest published analogue of our belief mechanism. PEBOL [3] keeps a per-item Beta posterior updated by LLM-extracted aspects and NLI scoring, reporting cold-start MRR@10 0.27 versus a 0.17 baseline. None of these reports full-profile accuracy at the EASE/RecVAE bar; each is a belief layer, not a certified tower.

**The set-encoder and cold-start-NP precedents (families i, iii).** EDDI/Partial-VAE [24] is the permutation-invariant set-encoder over an arbitrary observed subset with information-gain acquisition—exactly R2’s architecture, but demonstrated outside recommendation and without a full-profile CF result or an attribute/concept channel. TaNP [20] gives a neural-process, stochastic-process uncertainty over a value-carrying set input for cold-start—the R2+R4 shape, but not at full-profile SOTA nor channel-agnostic. These are cited as the mechanism precedents our design instantiates, not as rivals for the conjunction.

**The gap map.** Figure 1 places every system on two axes: *interview-compatibility* on the  $x$ -axis—a coarse ordinal capability count, the number of **R2–R5** marks a system satisfies in the master table ( $\checkmark=1$ ,  $\sim=0.5$ ; R1 is excluded because accuracy is the  $y$ -axis)—and *measured full-profile NDCG@10 on our ML-25M ruler* (R1) on the  $y$ -axis. The population is bimodal: the accuracy corner collapses onto  $x \leq 1.5$ , the interview-capable systems sit at  $x \geq 2$  with weak or non-existent full-profile accuracy, and no system bridges—the top-right, high compatibility and frontier accuracy, is empty.

## 4 Gap analysis

**The verdict.** No single published system satisfies all of  $\{\text{R1 SOTA-full-profile} \wedge \text{R2 any-length set} \wedge \text{R3 channel-agnostic via arbitrary embeddings} \wedge \text{R4 uncertainty-native} \wedge \text{R5 monotone}\}$ . The binding near-miss is Büyüç et al. [4]: it fails R1 (a co-trained item space, not a frozen certified RecVAE-class tower, and no full-profile benchmark) and gives only an approximate, “generally not Gaussian” posterior; every other candidate fails R3 or R4 outright. **We are aware of none** that occupies the conjunction—and we state it exactly that way, never as “there is none.”

### Per-near-miss failure analysis.

- **Büyüç et al. 2023** [4] — has R2 + R3 (items and soft attributes as directions) + R4 (Gaussian belief). *Fails R1*: the item space is co-trained with the belief model, not a certified full-profile CF tower, and accuracy is reported on their elicitation task, not the EASE/RecVAE bar. R3 is *attribute directions*, not arbitrary embeddings / open concepts / continuous query tokens. R5 is approximate (sampled EVOI over a non-Gaussian posterior). This is the central threat, and our wedge is exactly (a) frozen certified tower vs. co-trained, (b) exact conjugacy vs. approximate posterior, (c) arbitrary / open / continuous tokens vs. a fixed attribute set.
- **ConTS** [17] — R2 + R4 + a unified item/attribute action space. *Fails R3* (a fixed categorical attribute schema; no arbitrary embedding, no continuous or open token) and *R1* (a bandit over arms, not a full-profile encoder).
- **EDDI / Partial-VAE** [24] — R2 + R4 (information-gain) + set input. *Fails R1* (not a recommendation model / no CF SOTA) and *R3* (no attribute or concept channel as demonstrated).
- **RecVAE / EASE** [31, 33] — *R1 yes*. *Fail R3* (an item-only one-hot / indicator basis—cannot express a concept, a signed value, or a continuous direction) and *R4* (point estimate; RecVAE’s latent posterior is item-only and is not used as an actionable belief).
- **PEBOL** [3] — R4 (per-item Beta) + R3 (LLM aspects) + strong cold-start. *Fails R1* (no full-profile CF ranking) and its R4 is *per item*, not a shared user covariance; no monotonicity guarantee.
- **SASRec / BERT4Rec** [15, 35] — R2 (re-encode the sequence). *Fail R3, R4, R5* (item-only, order-sensitive, point estimate, non-monotone—a later token can reorder everything). Token input is trivially satisfied but is *not* value-carrying or mixed-type, so “interview = tokens” does not hold; and BERT4Rec’s fold-in advantage carries a replicability caveat [28].

**The three inexpressible legs.** Three input channels are *architecturally impossible* (not merely untuned) in the item-indicator basis of every R1-bar model: (i) graded/signed per-entity values (the basis has no sign and no magnitude—a “dislike” is not representable as a negative one-hot



without changing the likelihood), (ii) out-of-catalogue concepts (there is no column for an entity absent from the catalogue), and (iii) arbitrary continuous direction tokens. Folding these as *linear observations in a frozen latent* is the one leg of the conjunction we find nowhere in the literature.

**Pre-empted—cited, never claimed as first.** We must credit, and do not claim priority over: attribute-answers-as-directions plus a Bayesian fold [4]; unified item+attribute belief and value-of-information [4, 17]; conjugate-Gaussian folds over a frozen factorisation [36]; permutation-invariant set-encoders with information-gain acquisition [24]; value-carrying set-input cold-start recommenders [20]; the non-degradation-of-information principle [12, 5]; and the now-crowded “uncertainty-native recommender” area [6]. Our defensible novelty is only the *conjunction* of all five requirements in one frozen certified tower, plus the three inexpressible legs and the G0–G2 protocol.

## 5 The instrument

The design follows directly from the gap analysis: use the shallow frozen tower that already holds the accuracy corner (R1, and the flat-since-2019 frontier means no deep-net detour is warranted), and add the one belief mechanism with a principled monotonicity story—exact linear-Gaussian conjugacy—over its frozen latent (R4, R5), with a channel-agnostic linear-observation interface (R3). The tower half of this design is built and certified (§7); the belief layer is specified here at design level and its numbers are **[TODO: ]**.

### 5.1 Frozen shallow-SOTA tower

The substrate is a certified shallow recommender—a RecVAE-class amortised encoder or a closed-form autoencoder [31, 33]—trained once on the full catalogue and then *frozen*. It exposes (a) item factor rows  $\mathbf{Q}_i \in \mathbb{R}^d$ , (b) a decoder read-out  $s_i(\mathbf{u}) = \mathbf{Q}_i^\top \mathbf{u} + \beta_i$  with a learned bias  $\beta_i$ , and (c) the encoder that maps a full interaction set to a latent. Freezing is the wedge against Bıyık [4] (co-trained) and the guarantee that G0 (full strength preserved) is a property of the substrate, not of the belief layer. We deliberately do *not* extend the linear basis with concept columns (that stays in the item-indicator world and fails R3), nor learn the uncertainty head (no monotonicity story, and retraining risks G0). The tower is now *selected and certified*: the graded-native set-encoder T2′, trained once in its pre-registered configuration, scores 0.3482/0.2462 full/tail on the held-out test cohort—a paired-bootstrap tie with the 0.3540/0.2497 RecVAE bar—and reproduces the frozen decoder exactly at the empty interview (§7). It is this checkpoint that the belief layer below is placed over.

### 5.2 Analytic conjugate-Gaussian belief layer

Over the frozen latent we place a Gaussian belief on the user vector,  $\mathbf{u} \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ , and model each answer as a noisy linear read-out along the queried direction:

$$y_j = \boldsymbol{\phi}_j^\top \mathbf{u} + \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, \sigma_j^2). \quad (1)$$

Because the observation is linear-Gaussian, the posterior after  $\mathcal{E}_t$  is Gaussian in closed form; maintaining it in *precision* form (a precision accumulator) makes the any-length fold-in a single running sum:

$$\boldsymbol{\Lambda}_t = \boldsymbol{\Lambda}_0 + \sum_{j=1}^t \sigma_j^{-2} \boldsymbol{\phi}_j \boldsymbol{\phi}_j^\top, \quad \boldsymbol{\Lambda}_t \boldsymbol{\mu}_t = \boldsymbol{\Lambda}_0 \boldsymbol{\mu}_0 + \sum_{j=1}^t \sigma_j^{-2} y_j \boldsymbol{\phi}_j, \quad \boldsymbol{\Sigma}_t = \boldsymbol{\Lambda}_t^{-1}. \quad (2)$$

Equation (2) is exactly the mechanism BCIE [36] uses for critiques over a frozen factorisation; our contribution is doing it over a *certified full-profile SOTA* tower with channel-agnostic tokens, and certifying the G0–G2 protocol on it. Three properties fall out by construction: it is a permutation-invariant *set* operation defined for every  $t \geq 0$  (R2); each update adds a rank-1 term along  $\phi_{t+1}$ , so  $\Lambda_t$  only grows and  $\det \Sigma_t$  only shrinks (G1); and conjugate updates are monotone in *expected utility* [12, 5]. The last is why R5 is plausible, but—because the tower’s dot-product ranker does not act optimally under the belief—per-question NDCG monotonicity is *not* automatic and must be shown empirically (§8). **[TODO: Fix  $\Lambda_0, \mu_0$  (prior from the tower), the per-channel noise  $\sigma_j$ , and the confidence/temperature schedule; verify G0 holds at the belief mean on the full profile.]**

### 5.3 Channel-agnostic tokens as linear observations

Every channel supplies a direction  $\phi_j \in \mathbb{R}^d$  and a value  $y_j \in \mathbb{R}$  to Eq. (2); the update is identical across channels (R3):

- **Items.**  $\phi_j = \mathbf{Q}_i$  (the frozen item factor). A graded or signed answer sets  $y_j$ ; a dislike is a negative  $y_j$  along the same direction—representable here precisely because the observation model (1) has a sign and a magnitude, unlike a one-hot indicator.
- **Out-of-catalogue concepts.**  $\phi_j$  is a concept direction in the latent (e.g. a centroid of member factors, or a text-adaptor image of the concept), folded exactly like an item—no catalogue column required.
- **Entities / free text.** An embedding is projected into the latent by a fixed adapter and folded identically.
- **Continuous direction tokens.** A policy may emit any  $\phi_j \in \mathbb{R}^d$  directly (the continuous-action interface the companion chapter uses); the belief update does not distinguish it from a catalogue entity.

The signed/graded, out-of-catalogue, and continuous legs are the three channels inexpressible in the R1-bar basis (§4). **[TODO: Specify the concept-direction and text-adaptor constructions, the polarity/gradation encoding, and the acceptance tests (a like/dislike flip reverses the queried neighbourhood; a concept fold lifts its member items; monotone in  $t$ ).]**

## 6 The concept channel: derivation, lineage, and failure analysis

The out-of-catalogue concept leg is the most demanding of the three inexpressible channels—it is the one with no catalogue column, no interaction count, and no sign in the item-indicator basis—so we work it out in full. Three things must be made concrete: how a concept answer becomes an observation pair  $(\phi_c, y_c)$  for the belief update of Eq. (2); where each piece sits in the published record (so that we claim only the conjunction, never a component); and—because the honest history of this channel includes a self-inflicted regression—what a measured failure of our *own* evaluation harness taught us about which gate class the battery was missing. This section is written at derivation level; the concept-channel numbers below are measured on the internal ML-25M interview harness and are reported as such.

**Provenance convention for this section.** Because this section reports numbers from more than one build, we state once which stack each came from and mark the rest inline. *Accuracy* numbers (G0, the baseline table) are on the certified tower of §7. *Interview* numbers—every per-question

curve in this section and the graded-input curve of Figure 2—are measured on a pre-certification snapshot of the same architecture, and the concept-channel curves additionally depend on which *fold operator* was trained: an earlier recipe whose concept capture was 1–6%, and the current one at roughly 25%. Results that stand on the current fold are reported plainly; results measured through the earlier fold are marked **[RE-RUN QUEUED: ]** and their re-runs are queued. Nothing in this section is a certified instrument curve, and we do not present it as one.

## 6.1 A signed, exposure-corrected affinity estimator

A concept answer must supply the value  $y_c$  that Eq. (1) reads along the concept direction. We want that value to be *signed* (a user can dislike “horror,” not merely fail to like it), *popularity-corrected* (a concept made of blockbusters must not read as universal affinity), *volume-corrected* (a heavy watcher must not read as liking everything), and *confidence-bearing* (a concept the user has plausibly never been exposed to is a *refusal*, not a dislike). Two estimators, an implicit and an explicit one, combine to meet this.

Let  $\mathcal{I}_u$  be the user’s watched set,  $p_i$  the catalogue marginal of item  $i$ , and let concept  $c$  have member set  $M_c$ . Write  $n_c = |\mathcal{I}_u \cap M_c|$  for the user’s watched members and  $e_c = |\mathcal{I}_u| \cdot \pi_c$  for the number *expected* under the user’s volume times the concept’s popularity  $\pi_c = \sum_{i \in M_c} p_i$ . The implicit *selection* estimator is the add- $\frac{1}{2}$  log watch-lift

$$\text{SEL}(u, c) = \log_2 \frac{n_c + 0.5}{e_c + 0.5} \approx \text{PMI}(u, c), \quad (3)$$

which is precisely the *pointwise mutual information* between the user and the concept: the denominator’s  $\pi_c$  divides out concept popularity, and forming  $e_c$  from the user’s own volume divides out user activity, so SEL is positive for genuine affinity, near zero for popularity-explained co-occurrence, and *negative for avoidance*. Its normalised form is the NPMI,  $\text{NPMI} = \text{PMI}/(-\log_2 \hat{p}_{uc}) \in [-1, 1]$ , which tames PMI’s small-count inflation and yields a bounded, comparable score. The explicit *value* estimator is a support-shrunk signed rating residual,

$$\text{VAL}(u, c) = \frac{n_c}{n_c + \lambda} \bar{r}_{uc}^{\text{res}}, \quad \lambda = 3, \quad \bar{r}_{uc}^{\text{res}} = \frac{1}{n_{c, i \in \mathcal{I}_u \cap M_c}} \sum (r_{ui} - \bar{r}_i), \quad (4)$$

the mean rating the user gives the concept’s members *above each item’s own mean*, shrunk toward zero by the Beta-style factor  $n_c/(n_c + \lambda)$  so that a one-film opinion counts for little. Both estimators are signed and combine into a four-band answer: a *graded like* (SEL/VAL band above the neutral zone), a *neutral* (indistinguishable-from-exposure), a *graded dislike* (band below), and a *refusal* issued exactly when the expected support falls below a threshold,  $e_c < \tau$ , i.e. when the user has plausibly never been exposed to the concept and no answer should be forced.

**Lineage.** Every piece is established prior art, which is the point of stating it explicitly. The log-lift (3) is PMI/NPMI, the standard popularity-corrected association score. Reading a *magnitude* of implicit consumption as evidence is Hu–Koren–Volinsky’s implicit-confidence template [14]—except that HKV lets the count re-enter *only as a loss confidence weight, never as a sign*, which is exactly the assumption our signed estimator breaks. The principled account of why a non-watch is not a dislike is ExpoMF [18]: it separates a latent *exposure* variable (was the user even aware of the item) from *preference*, so that non-consumption is modelled as “not exposed” *or* “exposed-and-disliked”—the generative gold standard our refusal band ( $e_c < \tau$ ) and popularity-corrected  $e_c$  approximate cheaply and in closed form. The unbiased-learning route to the same correction is

attribute-level exposure propensity [29], which IPS-reweights implicit feedback by a per-attribute propensity; it is the heavier, variance-prone alternative to our moment estimator. Finally, calibrated recommendation [32] builds a user’s normalised genre *distribution*  $p(g \mid u)$ —the closest widely-used “concept profile,” but a distribution that sums to one and therefore encodes relative emphasis only: it *cannot express dislike*, which is precisely the expressive gap  $\text{SEL} < 0$  fills.

**Why the implicit channel, quantified.** On an informativeness decomposition (a per-concept linear fit of the oracle concept affinity onto each candidate answer,  $R^2$ ), the implicit watch-lift dominates: SEL explains 0.376 of the affinity variance, the explicit rating residual VAL a further +0.042 (SEL+VAL 0.398), while an LLM-elicited ordinal answer explains only 0.017 (an internal decomposition)—implicit outperforms the explicit LLM answer by roughly  $22\times$ . Because SEL is built from raw watch counts and item marginals, it uses no recommender geometry, so this signal is *non-circular by construction* (it even beats a model-geometry projection of the same residuals). This is the evidence that the concept channel’s value is worth carrying signed rather than as a mere membership indicator; the decomposition is an informativeness measurement, not an interview curve, and is reported as such.

## 6.2 The fold operator: whitened directions and a gated fold-to-point

The direction  $\phi_c$  must point where the concept’s taste lives, not where popularity lives. A raw member centroid fails this: a shared popularity component (random concept pairs have cosine  $\approx 0.5$ ) masks the genre geometry. Removing it—centre the member factors and strip the top-1 principal component (the popularity axis), the all-but-the-top whitening of Mu and Viswanath [25]—turns the centroid into a usable direction: concept-membership discrimination through the frozen decoder rises from AUC 0.44 (raw, at chance) to 0.94 (whitened). We therefore take  $\phi_c$  to be the *whitened member centroid*, IDF-weighted by  $1/\log(1 + |M_c|)$  so broad genres do not dominate. Whitening is a fix, not a contribution, and is cited as such.

The operator that consumes  $(\phi_c, y_c)$  is where the design decision lives, and our own history is a clean ablation between two families. An *additive-union* operator scores each item by summing concept contributions on a popularity floor,  $s_i = \beta_i + \sum_c w_c \langle \mathbf{W}_d \mathbf{Q}_i, \phi_c \rangle$ . It has the right monotonicity floor (an empty interview scores exactly  $\beta_i$ ) but the wrong composition: it scores the *union* of directions, so correlated sibling concepts (“dark,” “gritty,” “noir”) re-add the same acclaim axis and over-count. Measured against a no-answer intercept of 0.1279 full NDCG@10, the additive channel peaks early (about +0.011 at two answers) and then *craters* as answers accumulate, reaching 0.0991 at eight answers—below the intercept. A *trained gated fold-to-point* operator fixes this by folding the answers into the latent itself and letting a frozen decoder apply the full collaborative (intersection) structure:

$$\mathbf{u} \leftarrow \mathbf{u} + \sigma(g([\mathbf{u}; \phi_c; y_c])) \odot h([\mathbf{u}; \phi_c; y_c]), \quad (5)$$

with the gate  $g$  and the delta MLP  $h$  *zero-initialised* (so at initialisation  $\mathbf{u}$  is unchanged and G0 survives) and the accumulated shift norm-capped per user. This operator is monotone in the number of answers to eight (concepts-only 0.2091/0.1222 full/tail at eight answers, up from the 0.1717 cold point) and, crucially, *redundancy-robust*: feeding it eight *correlated* top-SEL concepts still reaches 0.2219 full, at or above its four-answer level, where the additive-union operator degraded. **[RE-RUN QUEUED: operator-family magnitudes above are from the earlier fold recipe; the additive-vs-gated verdict is what carries forward, and the levels are re-measured with the fixed fold]** The gated fold is the concept operator we adopt; Eq. (5) is the concept-channel instantiation of the linear-observation fold, with the belief covariance maintained analytically as in §5.

**Lineage.** The skeleton—an attribute as a latent direction, folded by a Bayesian belief update over a frozen factor space—is well-trodden, and we claim none of it in isolation. Büyük et al. [4] treat a soft attribute as a concept-activation-vector direction and fold signed answers into a Gaussian belief; M&Ms-VAE folds a rating-posterior and an attribute-posterior over one latent by a product-of-experts merge [1]; its successor adds an independent *signed* critique axis through learned positive/negative conditioning tokens, and reports that negative critiquing saturates after about five turns [2]—a concrete ceiling for any signed-attribute channel, consistent with our own front-loaded concept curves. BK-VAE folds keyphrase-activation-vector directions by a Bayesian critiquing step [23]; the gate/MLP of Eq. (5) is a per-token generalisation of FiLM conditioning [27], which a single global affine cannot do because it would apply the same modulation to item and concept tokens alike. And the decision to *freeze* the tower rather than co-train a concept head is a direct reading of the cautionary result that a bolt-on critique head costs recommendation accuracy [22]. What none of these holds jointly—and what we are aware of no system holding—is the conjunction this channel occupies: *continuous, out-of-catalogue concept directions* (a whitened centroid, or a text-adaptor image, with no per-attribute probe to train), driven by a *signed, exposure-corrected implicit affinity* (Eqs. (3)–(4)), folded over a *frozen, certified full-profile SOTA tower*. The components are borrowed; only their conjunction is claimed.

### 6.3 Failure analysis: a positive-only clip and the deployment-currency gate

The honest history of this channel includes a regression we introduced ourselves, and it is worth recording because it exposes a whole *class* of gate the battery originally lacked. When the concept curves were first built, the evaluation harness clipped every folded concept value to  $[0.25, 1]$ —the like-levels only—silently discarding the negative half of the estimator that §6.1 works so hard to produce. Selection kept only positive-lift concepts and floored the survivors at a weak positive;  $SEL < 0$  was *computed and then thrown away*.

The consequence only became visible in the right currency. Per-answer curves that privilege each user’s own top-SEL concepts looked fine, but those are unknowable before the interview. Measured in *deployment currency*—a realizable, generic question bank with a per-question budget, where most answers down a fixed bank are weak positives—the accumulated weak-positive “poison” dragged the ranking *below the no-answer intercept*: concepts-only full NDCG@10 declined from 0.1304 to 0.1160 by the sixteenth question, under the 0.1279 intercept. The channel was structurally unable to say “no,” even though the fold operator of Eq. (5) is *sign-capable*—a flip test on it moves NDCG by  $-0.201$  with a clean CI. That is the crux: **a capability gate is vacuous if the answer model never exercises the capability**. The sign-flip gate (G3) passed on a channel that, as actually driven in deployment, never emitted a negative answer at all.

We close this with a pre-registered, eval-only triple gate (recorded as the signed-SEL gate specification) that a retrain must clear before the signed answer is adopted:

1. **Recovery.** Signed answers routed through the sign-capable operator recover  $+0.0076$  full NDCG@10 at the sixteenth question with a CI excluding zero, and about  $1.5\times$  the tail—turning the deployment decline into a gain.
2. **Counterfeit control (taste, not artifact).** An adversarial review flagged that “dislike from absence” could be a  $\text{volume}\times\text{popularity}$  artifact; a counterfeit answer built only from user volume and concept popularity reproduces *zero* of the recovery, so the gain is taste, not an exposure artifact.
3. **Out-of-curriculum control.** A meh-only control shows the residual late decline is out-of-curriculum extrapolation—the operator was trained on interviews of up to eight answers and

asked sixteen—rather than a sign defect, isolating the two failure modes.

Because “dislike from absence” can leak user volume into the belief, we pair the gate with a volume-invariance acceptance criterion, backed by a probe: the recoverability of log watch-volume from the belief latent ( $R^2$  of a linear read-out) rises from 0.025 to 0.249 once signed answers are folded, so a signed-answer build is accepted only if that leak is held down (a volume-orthogonalised fold, or an invariance penalty).

**The lesson for the battery.** The outcome-gate false positives of §2 have a sibling among the *semantic* gates. A capability gate such as G3’s sign-flip certifies that the operator *can* negate; it says nothing about whether the operator, driven by a realistic answer model down a realistic question bank, ever *does*. The miss is closed only by a *deployment-currency* gate—evaluate the channel on a fixed, user-independent question bank under a per-question budget (the G9 ingestion-isolation setting, scored at the endpoint), where an answer model that never exercises the negative half is penalised by the same declining curve that exposed the clip. We therefore add the deployment-currency endpoint as the acceptance form of the concept channel: not “can the operator express dislike” but “does the realizable interview stay above the no-answer intercept at budget.”

## 6.4 Results: the signed retrain, and where the concept channel wins

The signed retrain of §6.3 has since landed, and we ran the two channels—concepts and items—through *one* deployment-currency harness (10,000 held-out cold users, per-question budget, every question spending budget whether answered or not, scored on full *and* tail NDCG@10 against the no-answer intercept 0.1279/0.0192). The results are reported here on the internal ML-25M interview harness; the certified instrument curves remain [TODO: ] (§7). They tell a consistent, and deliberately un-triumphal, story: the signed answer repairs the regression, but the concept channel’s value is *shaped*—it owns the opening of an interview and cedes the long tail of it to items—and its largest prize is one we do not yet capture.

**The signed answer turns the deployment decline into a gain.** Re-running the pre-registered triple gate on the *retrained* model (not merely the eval-only fold of §6.3) clears all three taste criteria. Signed answers beat the positive-only clip by +0.0104 full NDCG@10 at the sixteenth question (CI [+0.0075, +0.0134]): the clip-up curve sinks below the intercept by the eighth question and stays there (0.1268 at  $q=16$ ), while the signed curve holds above it (0.1372). The counterfeit control—an answer built only from user volume and concept popularity—reproduces *none* of the gain (it declines, −0.045 at  $q=16$ ), so the signal is taste, not a volume×popularity artifact; and permuting the answer values across concepts collapses the gain (+0.0335 CI-excluding-zero relative to the permutation), so it is the answer *values* that carry it, not fold cardinality. *One acceptance criterion remains unmet, and we record it plainly:* the volume-invariance gate still fails. The recoverability of log watch-volume from the belief latent is  $R^2$  0.249 after the retrain, against a pre-registered bar of  $\leq 0.05$ —the per-user negative-channel cap did not hold the leak down. We treat the signed concept answer as *directionally supported but not yet certified*: the counterfeit-clean result is the waiver-consideration evidence (the leaked quantity is user volume, but the ranking gain provably is not a volume meter), and the fold’s acceptance is pending either a volume-orthogonalised variant or an author ruling on the waiver. [RE-RUN QUEUED: the triple-gate magnitudes above are measured through the earlier fold recipe; the three verdicts (recovery, counterfeit-clean, values-carry-it) are what the gate certifies, and the deltas are re-measured on the fixed fold]



**Both channels carry an interview, and the concept channel is strongest exactly where the item channel is weakest.** The instrument’s claim on the elicitation side is deliberately modest: not that concepts beat items, but that *both* channels fold and both move the ranking, which is what makes the instrument usable for measuring policies over a mixed question bank. Under a matched question budget—a realizable concept ask against a popularity-ranked item ask, both scored on the same cold cohort against the 0.1279/0.0192 intercept—the two channels separate cleanly by regime (full/tail NDCG@10):

| questions $k$ | 1                  | 2                  | 4                  | 8                  |
|---------------|--------------------|--------------------|--------------------|--------------------|
| concept ask   | <b>.1591/.0638</b> | <b>.1803/.0861</b> | <b>.1972/.1050</b> | .2107/.1210        |
| item ask      | .1374/.0392        | .1493/.0478        | .1839/.0648        | <b>.2274/.0839</b> |

The concept channel leads on *full* accuracy through  $k=4$  and dominates the *tail* at every budget measured—at  $k=8$  the tail gap is .1210 versus .0839, about +44%—while items overtake on full accuracy only at  $k=8$ . The mechanism is the coarse-to-fine intuition made concrete: a concept question splits a cold population along a taste axis in one move, where a single item barely moves the belief; and because a concept direction is not anchored to any one popular title, its gains land disproportionately off the head, which is precisely the tail. Two honest caveats attach. The comparison is a realizable concept ask against a *popularity* item ask, which is the item channel’s strongest simple strategy but not its ceiling; and items do win full accuracy by  $k=8$ , so this is a statement about *shape*, not a claim that one channel dominates. What the instrument chapter needs from this is only the weaker conclusion, which is unambiguous: an interview conducted entirely in out-of-catalogue concepts—a channel no R1-bar baseline can express at all—moves a frontier-accuracy recommender substantially and monotonically off its cold intercept.

**A retraction: “items win the long interview” was an artifact of a broken fold.** An earlier version of this section reported the opposite shape past  $q=4$ —that popularity items pulled away from a concept plateau near 0.147 from the eighth question onward—from a strategy×channel suite run on a superseded concept fold. That fold was trained under a recipe whose concept-capture rate was 1–6%; repairing the recipe (a signed four-band answer plus a curriculum over the number of concepts and dropped members) raised capture to roughly 25% and inverted the long-interview reading, as the table above shows. We record the retraction rather than quietly restating the numbers, because it is the same lesson the rest of this section is about: a channel’s measured ceiling is a property of the operator it is measured through, and a weak operator manufactures a confident negative result. The full strategy×channel picture—question-bank strategies, and the best *static* question sequence obtained by greedy NDCG maximisation over an items-only, concepts-only, and combined bank—is being re-measured on the fixed fold with construction and evaluation on disjoint cohorts. **[RE-RUN QUEUED: strategy×channel suite and the best-static greedy sequences, fixed fold, build on the 10k validation cohort / evaluate on the disjoint 10k test cohort; figure to be inserted here]**

**On the adaptive headroom.** The earlier suite also reported a large gap between a privileged oracle selector and every realizable static arm, with a naive greedy selector recovering only a low-single-digit fraction of it. Those magnitudes were measured through the same superseded fold and are withdrawn pending the re-run (**[RE-RUN QUEUED: oracle–static gap and greedy closure on the fixed fold]**). The *direction* survives on evidence independent of the fold: a one-level adaptive probe on the fixed stack—branch on the first answer, then choose the second question per

branch, selected and evaluated out-of-sample on disjoint halves—pays on the item channel with a CI excluding zero (+0.0015 full, CI [+0.0007, +0.0024]; +0.0016 tail, CI [+0.0008, +0.0024]), and is positive but not significant on the concept channel at one level. The effect is small because a one-level preview is not a policy; that is the point. Establishing that adaptivity pays at all, on an instrument whose ingestion is certified, is what this chapter owes the companion method chapter, whose job is to close the gap—the decision-tree acquisition line of Golbandi et al. [11] on *this* instrument and harness. The instrument chapter’s job is to make the gap *measurable*; closing it is the method chapter’s.

**Training concepts into the tower is item-safe but buys nothing.** We evaluated two homes for the signed concept fold: a trained fold-to-point on the *frozen* tower (call it the frozen-fold), and concept tokens trained *into* the tower itself (the tokens-in-tower variant). Both preserve full-profile strength, as G0 demands. The frozen-fold inherits the tower’s G0 by construction (a zero-initialised fold is bit-identical on the full profile). The tokens-in-tower variant is retrained and therefore must re-earn G0, and it does: full/tail NDCG@10 0.3487/0.2471 on the test users, a paired-bootstrap tie with the 0.3540/0.2497 RecVAE bar (difference  $-0.0053$ , within the 0.0070 CI), with cold-start accuracy if anything marginally higher. So *training concepts into the tower costs essentially nothing on items*. It also, however, *gains* essentially nothing on concepts, and it is worse where it matters most. On per-answer concept curves the frozen-fold is higher at eight answers (0.1630/0.0639 versus 0.1566/0.0577), and in deployment currency the difference is decisive: the frozen-fold stays above the intercept through sixteen questions (0.1379 at  $q=16$ ), whereas the tokens-in-tower variant *craters* to 0.1052, below the no-answer intercept, as the long tail of the fixed bank floods it with weak-support concepts. We therefore carry the frozen-fold as the concept operator of record; the tokens-in-tower arm is item-safe but does not earn its place. The G0 comparison above is a genuine tie on the certified ruler; the concept curves that decide *between* the two homes were measured through the earlier fold, so the ranking of the two homes is re-checked on the fixed fold before it is final. **[RE-RUN QUEUED: frozen-fold vs. tokens-in-tower concept curves on the fixed fold]** The remaining open item is the volume-leak waiver above, which fixes the final certified operating point.

**The methods lesson, restated as a battery amendment.** The concept channel’s near-miss is the sharpest argument in this chapter for *why the battery has the shape it does*. A capability gate (G3’s sign-flip) passed on a fold that, as actually driven down a realistic question bank, never emitted a negative answer—the answer model, not the operator, was sign-blind, and the operator’s certified capability was vacuous in deployment. Only the deployment-currency evaluation—a fixed, user-independent bank under a per-question budget, scored at the endpoint (the G9 ingestion-isolation setting)—exposed it, because there the declining curve penalises an answer model that never exercises the negative half. This is a general failure class, not a concept-specific accident: *a gate on what an operator can do must be paired with a gate on what the deployed system does*. We record the deployment-currency endpoint as the concept channel’s acceptance form, and fold the signed-retrain and volume-leak numbers into the G3/G9 rows once the certified tower is trained (§7).

## 7 Experimental design

### 7.1 Baseline bank

The baseline bank (Table 4) is three tiers: classic floors that every curve must clear (Dacrema et al. [10] demands them), the full-profile SOTA bar that decides G0, and the elicitation-compatible



rivals that decide G1/G2. Reported anchors are on the systems’ *own* splits and are not comparable to ours until bridged (§7.2); we print them only to show what “ties SOTA” will mean.

## 7.2 Evaluation protocol

The protocol items below instantiate the headline outcome gates (the certification bridge is C1; G0 and G2 as named); they are the cheap, always-run subset. The remaining per-build gates (G3–G9) and the load-bearing answer-model-transfer certification (C2), which close the outcome-gate false positives, are specified in full in the acceptance battery (Table 1) and its design-side document, and each headline result is reported against them.

**(i) Certification bridge (C1).** Our ML-25M ruler and the published bar are on *different* metrics and splits, and we do not conflate them. Concretely, our paper-config RecVAE realisation reaches full/tail NDCG@10 0.3540/0.2497 on the canonical ML-25M split—whereas the published EASE/RecVAE numbers are NDCG@100 on the ML-20M strong-generalisation split. These are not comparable, and no “ties SOTA” claim is made until the bridge is run. The bridge has two steps: (1) *reproduce* the published EASE and Mult-VAE/RecVAE numbers on the ML-20M Liang split (NDCG@100, Recall@20/50) in our own code, snapping to the reported values exactly [33, 19, 31]; (2) *port* the certified tower to the ML-25M harness and report full/tail NDCG@10 there. We keep ML-20M in the loop despite being older because the Liang 2018 split is a protocol lock-in the whole full-profile literature reports on; ML-25M is the newer corpus and, crucially, carries the genome tags the concept channel needs, so both datasets are required—one for comparability, one for the channels.

The first step is complete. On the ML-20M Liang strong-generalisation split our split statistics reproduce the published counts exactly (9,990,682 events / 116,677 train users / 20,108 items), and both anchor implementations **snap to their published values**: EASE at NDCG@100 0.4203 vs. the reported 0.420 [33], and RecVAE at 0.4425 vs. the reported 0.442, with Recall@20 (0.4144 vs. 0.414) and Recall@50 (0.5525 vs. 0.553) matching as well [31]. One closed-form and one neural reproduction to the third decimal certify that our split construction, metric implementation, and both implementation families are faithful to the published protocol—every ML-25M baseline number below therefore comes from certified code.

The other two amortised anchors are reported with the same candour, because a bridge that only ever passes is not a bridge. **Mult-VAE** reaches NDCG@100 0.4223 against the published 0.426 [19] (−0.0037; Recall@20 0.3928 vs. 0.395, Recall@50 0.5341 vs. 0.537)—a *near-snap*, close enough to admit as a bridged Tier-2 anchor but not a third-decimal reproduction, and we label it that way rather than round it into agreement. **Mult-DAE** *fails*: 0.3969 against the published 0.419 (−0.022, with Recall@20 and @50 short by a comparable margin). We do not know whether the gap is a training-recipe detail or something deeper, and rather than carry an uncertified implementation we **drop Mult-DAE** from the baseline bank entirely—it earns no row on the ML-25M ruler. This is the bridge doing its job: the same rule that licenses the EASE and RecVAE numbers disqualifies the DAE one. We note for completeness that our iALS lands at 0.358 against the published WMF anchor of 0.386, so the iALS row throughout should be read as a *floor* for that model family rather than a tuned reproduction of it.

**(ii) G0 — full-profile on ML-25M, items-only.** On the baselines’ native input (items only), evaluate full and tail NDCG@10 on the full ML-25M catalogue with a paired per-user bootstrap CI, confirming the instrument’s belief mean reproduces the frozen tower and ties EASE/RecVAE on the shared harness. The item-only baselines are measured (Table 5).

**A properly-trained RecVAE edges the closed-form frontier, and that is the bar.** On our canonical ML-25M ruler our in-house paper-config RecVAE attains 0.3540 full / 0.2497 tail, the nominally strongest measured system—ahead of the closed-form leaders EASE (0.3476/0.2441) and EDLAE (0.3433/0.2395). The RecVAE-over-EASE margin is +0.0064 full, about 1.8 unpaired standard errors; we therefore state it as a *nominal* lead with a paired per-user bootstrap CI pending, not a settled win. All rows share the same split, cohort, tail definition, and NDCG@10 routine (verified by code inspection). The point of interest for the instrument is that on this ruler a properly-trained amortised VAE—not a closed-form model—sits at the top of the frontier; the instrument’s R1 obligation is precisely to *tie this bar*, reproducing RecVAE-class full/tail accuracy through the frozen tower and then preserving it under the belief layer at G0. We report the whole frontier rather than excluding the stronger baselines.

**G0 is passed: the interview tower ties the bar, and is exactly the frozen decoder when the interview is empty.** One certified training run of the graded-native set-encoder tower T2’—a single run of the pre-registered winning configuration, selected on validation and scored once on the held-out test cohort—reaches full/tail NDCG@10 **0.3482/0.2462** against the RecVAE bar’s 0.3540/0.2497. The difference is −0.0057 full with a paired per-user bootstrap CI of 0.0069, so the two *tie* in the sense of Definition 1: the tower pays no measurable full-profile price for being interviewable. Two properties of that number matter more than its size. First, it is on the *same* ruler as every other row of Table 5—same split, cohort, tail mask, and NDCG routine—so it is a tie in the strict sense, not a bridged one. Second, the G0 *identity* half holds exactly and not merely statistically: with an empty evidence set the tower emits a zero latent ( $|\mu_0| = 0$  to numerical precision) and its scores are a rank-identical image of the frozen decoder’s learned bias (Spearman 1.0000), which is what licenses reading every later interview curve as evidence about *ingestion* rather than about a shifted baseline. The cold intercept of every curve in this chapter, 0.1279/0.0192, is that same non-personalised read-out. Accuracy is therefore no longer the open question for this instrument; what remains open is whether the belief layer preserves the tie once it is placed over the frozen latent, which is **[TODO: ]** until the belief layer is fitted (§5).

**(iii) G2 — per-question curves.** Cold-start per-question NDCG@10 curves (full and tail) versus a random-question control. Items are foldable by all baselines; mixed channels (concepts, signed values, continuous tokens) are foldable only by our instrument, which is the point of the comparison—the rivals cap out at what an item-only interview can carry. Symmetric access is enforced where a baseline *can* fold a channel; where it structurally cannot, that is reported as an architecture limit, not hidden. **[TODO: G2 curves for items-only (all methods) and mixed-channel (ours), full & tail, with CIs and the random control; G1 posterior-shrinkage diagnostic.]**

**(iv) Concept-as-pseudo-item ablation.** To pre-empt “just add genre columns,” we fold a concept into EASE/RecVAE as the centroid-of-members pseudo-item and compare against our concept channel. The expected finding is that a pseudo-item column is a lossy special case (it cannot carry sign, magnitude, or an out-of-catalogue direction), but we run it rather than assert it. **[TODO: Ablation table: concept-as-pseudo-item fold into EASE/RecVAE vs. our concept channel, full & tail.]**

**Metrics.** Full *and* tail NDCG@10 on ML-25M (the project reporting rule, scored with the decoder’s learned bias, not a log-count popularity floor), plus NDCG@100 (and Recall@20/50) on the ML-20M bridge split for comparability with the published anchors. Every headline delta carries a

paired per-user bootstrap CI on one shared harness. The G0 and baseline cells are measured; the G1/G2 cells are **[TODO: ]** pending the belief layer.

## 8 Limitations and threats to validity

- **Metric non-comparability until the bridge is run.** Our frontier number 0.3540/0.2497 is NDCG@10 on ML-25M; the published bar is NDCG@100 on ML-20M. Until §7.2 snaps our code to the published numbers (done for EASE/RecVAE, §7.2), *no* “ties SOTA” claim across the two rulers is licensed—the ML-25M and ML-20M numbers live on different rulers.
- **Monotonicity is empirical, not a theorem.** Conjugate-Gaussian updates are monotone in expected utility [12, 5] *only in expectation*, and only if the ranker acts optimally under the belief—which a fixed dot-product / NDCG ranker does not. R5 (and G2’s rising curve) is therefore a *measured* property with a per-question CI, not a guarantee; a truthful answer can in principle still lower NDCG at a given step, and we report any such step rather than suppress it.
- **Simulator-based evaluation pending a human study.** G1/G2 are measured under a stated answer model (a simulator), which shares structure with the instrument’s own latent; this risks circularity (the answer, the belief, and the metric are all functions of the same space). We add answer-noise ablations and a simulator-alignment limitation to every gate; the battery discharges this through certification C2 (the G2 gain must survive a held-out behavioural answer model plus injected noise, with a geometry-answer firewall) and treats the  $\geq 1$  human round-trip study (C3) as the load-bearing external validation—not yet run.
- **BERT4Rec replicability.** The sequence baseline’s fold-in advantage is subject to a documented replicability caveat [28]; we tune it to the published protocol and flag any residual gap rather than treat a single run as authoritative.
- **Graph-filter admissibility.** GF-CF/BSPM/Turbo-CF [30, 7, 26] do not publish the ML-20M strong-generalisation NDCG; they enter Tier 2 only if re-run on our split, and are otherwise reported as an accuracy ceiling on other datasets, not as a bridged tie.
- **Absence of evidence.** The conjunction claim is “we are aware of none,” bounded by our survey; a set-encoder / deep-set recommender that hits RecVAE-class full-profile accuracy *and* takes value-carrying mixed-type tokens would partly pre-empt R1 $\wedge$ R2 $\wedge$ R3. Our searches found no such reported system, which is absence of evidence, not proof of absence.

## References

- [1] Diego Antognini and Boi Faltings. Fast multi-step critiquing for VAE-based recommender systems. In *Proceedings of the 15th ACM Conference on Recommender Systems (RecSys)*, 2021. arXiv:2105.00774.
- [2] Diego Antognini and Boi Faltings. Positive and negative critiquing for VAE-based recommenders. *arXiv preprint arXiv:2204.02162*, 2022.
- [3] David Eric Austin, Anton Korikov, Armin Toroghi, and Scott Sanner. Bayesian optimization with LLM-based acquisition functions for natural language preference elicitation. In *Proceedings of the 18th ACM Conference on Recommender Systems (RecSys)*, 2024.

- [4] Erdem Bıyık, Fan Talati, and Dorsa Sadigh. Preference elicitation with soft attributes in interactive recommendation. *arXiv preprint arXiv:2311.02085*, 2023.
- [5] David Blackwell. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24(2):265–272, 1953.
- [6] Radin Cheraghi, Amir Mohammad Mahfoozi, Sepehr Zolfaghari, Mohammadshayan Shabani, Maryam Ramezani, and Hamid R. Rabiee. Epistemic uncertainty-aware recommendation systems via bayesian deep ensemble learning. *arXiv preprint arXiv:2504.10753*, 2025.
- [7] Jeongwhan Choi, Seoyoung Hong, Noseong Park, and Sung-Bae Cho. Blurring-sharpening process models for collaborative filtering. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, pages 1096–1106, 2023.
- [8] Paolo Cremonesi, Yehuda Koren, and Roberto Turrin. Performance of recommender algorithms on top-n recommendation tasks. In *Proceedings of the 4th ACM Conference on Recommender Systems (RecSys)*, pages 39–46, 2010.
- [9] Yang Deng, Yaliang Li, Fei Sun, Bolin Ding, and Wai Lam. Unified conversational recommendation policy learning via graph-based reinforcement learning. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, pages 1431–1441, 2021.
- [10] Maurizio Ferrari Dacrema, Paolo Cremonesi, and Dietmar Jannach. Are we really making much progress? a worrying analysis of recent neural recommendation approaches. In *Proceedings of the 13th ACM Conference on Recommender Systems (RecSys)*, pages 101–109, 2019.
- [11] Nadav Golbandi, Yehuda Koren, and Ronny Lempel. Adaptive bootstrapping of recommender systems using decision trees. In *Proceedings of the Fourth ACM International Conference on Web Search and Data Mining (WSDM)*, pages 595–604, 2011.
- [12] I. J. Good. On the principle of total evidence. *The British Journal for the Philosophy of Science*, 17(4):319–321, 1967.
- [13] Zhankui He, Zhouhang Xie, Rahul Jha, Harald Steck, Dawen Liang, Yesu Feng, Bodhisattwa Prasad Majumder, Nathan Kallus, and Julian McAuley. Large language models as zero-shot conversational recommenders. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management (CIKM)*, pages 720–730, 2023.
- [14] Yifan Hu, Yehuda Koren, and Chris Volinsky. Collaborative filtering for implicit feedback datasets. In *Proceedings of the IEEE International Conference on Data Mining (ICDM)*, pages 263–272, 2008.
- [15] Wang-Cheng Kang and Julian McAuley. Self-attentive sequential recommendation. In *Proceedings of the IEEE International Conference on Data Mining (ICDM)*, pages 197–206, 2018.
- [16] Belinda Z Li, Alex Tamkin, Noah Goodman, and Jacob Andreas. Eliciting human preferences with language models. *arXiv preprint arXiv:2310.11589*, 2023.
- [17] Shijun Li, Wenqiang Lei, Qingyun Wu, Xiangnan He, Peng Jiang, and Tat-Seng Chua. Seamlessly unifying attributes and items: Conversational recommendation for cold-start users. *ACM Transactions on Information Systems (TOIS)*, 39(4):1–29, 2021.

- [18] Dawen Liang, Laurent Charlin, James McInerney, and David M. Blei. Modeling user exposure in recommendation. In *Proceedings of the 25th International Conference on World Wide Web (WWW)*, pages 951–961, 2016.
- [19] Dawen Liang, Rahul G. Krishnan, Matthew D. Hoffman, and Tony Jebara. Variational autoencoders for collaborative filtering. In *Proceedings of the World Wide Web Conference (WWW)*, pages 689–698, 2018.
- [20] Xixun Lin, Jia Wu, Chuan Zhou, Shirui Pan, Yanan Cao, and Bin Wang. Task-adaptive neural process for user cold-start recommendation. In *Proceedings of the Web Conference (WWW)*, pages 1306–1316, 2021. arXiv:2103.06137.
- [21] Nathan N. Liu, Xiangrui Meng, Chao Liu, and Qiang Yang. Wisdom of the better few: cold start recommendation via representative based rating elicitation. In *Proceedings of the 5th ACM Conference on Recommender Systems (RecSys)*, pages 37–44, 2011.
- [22] Kai Luo, Scott Sanner, Ga Wu, Hanze Li, and Hojin Yang. Deep language-based critiquing for recommender systems. In *Proceedings of the 13th ACM Conference on Recommender Systems (RecSys)*, pages 137–145, 2019.
- [23] Kai Luo, Hojin Yang, Ga Wu, and Scott Sanner. Deep critiquing for VAE-based recommender systems with keyphrase activation vectors. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, 2021.
- [24] Chao Ma, Sebastian Tschiatschek, Konstantina Palla, José Miguel Hernández-Lobato, Sebastian Nowozin, and Cheng Zhang. EDDI: Efficient dynamic discovery of high-value information with partial VAE. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, pages 4234–4243, 2019.
- [25] Jiaqi Mu and Pramod Viswanath. All-but-the-top: Simple and effective postprocessing for word representations. In *Proceedings of the 6th International Conference on Learning Representations (ICLR)*, 2018. arXiv:1702.01417.
- [26] Jin-Duk Park, Yong-Min Gim, U Kang, and Won-Yong Shin. Turbo-CF: Matrix decomposition-free graph filtering for fast recommendation. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, 2024. arXiv:2404.14243.
- [27] Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville. FiLM: Visual reasoning with a general conditioning layer. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018. arXiv:1709.07871.
- [28] Aleksandr Petrov and Craig Macdonald. A systematic review and replicability study of BERT4Rec for sequential recommendation. In *Proceedings of the 16th ACM Conference on Recommender Systems (RecSys)*, pages 436–447, 2022.
- [29] Zhen Qin, Suming J. Chen, Donald Metzler, Yongwoo Noh, Jingzheng Qin, and Xuanhui Wang. Attribute-based propensity for unbiased learning in recommender systems: Algorithm and case studies. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pages 2359–2367, 2020. WSDM/KDD line; Google unbiased-LTR.

- [30] Yifei Shen, Yongji Wu, Yao Zhang, Caihua Shan, Jiliang Zhang, Khaled B. Letaief, and Dongsheng Li. How powerful is graph convolution for recommendation? In *Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM)*, pages 1619–1629, 2021.
- [31] Ilya Shenbin, Anton Alekseev, Elena Tutubalina, Valentin Malykh, and Sergey I. Nikolenko. RecVAE: A new variational autoencoder for top-N recommendations with implicit feedback. In *Proceedings of the 13th ACM International Conference on Web Search and Data Mining (WSDM)*, pages 528–536, 2020. arXiv:1912.11160.
- [32] Harald Steck. Calibrated recommendations. In *Proceedings of the 12th ACM Conference on Recommender Systems (RecSys)*, pages 154–162, 2018.
- [33] Harald Steck. Embarrassingly shallow autoencoders for sparse data. In *Proceedings of the World Wide Web Conference (WWW)*, pages 3251–3257, 2019.
- [34] Harald Steck. Autoencoders that don’t overfit towards the identity. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, 2020.
- [35] Fei Sun, Jun Liu, Jian Wu, Changhua Pei, Xiao Lin, Wenwu Ou, and Peng Jiang. BERT4Rec: Sequential recommendation with bidirectional encoder representations from transformer. In *Proceedings of the 28th ACM International Conference on Information and Knowledge Management (CIKM)*, pages 1441–1450, 2019.
- [36] Armin Toroghi and Scott Sanner. Bayesian critique-improve-explain: Interactive recommendation with an uncertain latent-factor model. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, 2023.
- [37] Xiaoxue Zhao, Weinan Zhang, and Jun Wang. Interactive collaborative filtering. In *Proceedings of the 22nd ACM international conference on Information & Knowledge Management*, pages 1411–1420, 2013.
- [38] Ke Zhou, Shuang-Hong Yang, and Hongyuan Zha. Functional matrix factorizations for cold-start recommendation. In *Proceedings of the 34th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 315–324, 2011.

| Gate                                            | One-line criterion                                                                                          | Failure class it excludes                         |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <i>Per-build gates (every build; all cheap)</i> |                                                                                                             |                                                   |
| G0                                              | Belief mean reproduces the frozen tower’s full/tail NDCG@10 (tie or bit-identity).                          | Full-profile regression from the belief layer.    |
| G1                                              | $\Sigma$ shrinks in expectation over confident answers (refusals exempt), directionally, calibrated to NLL. | Unusable / uncalibrated posterior.                |
| G2                                              | Cold full/tail curve beats random by AUC-at-budget and strictly rises; one true answer helps every channel. | Flat-model and out-of-envelope artifacts.         |
| G3                                              | Sign-flip lowers NDCG; value-neutralisation costs NDCG; quality monotone in intensity.                      | Sign-blind and intensity-blind folds.             |
| G4                                              | Precisions order refuse < vague < know-well; collapsing levels loses; a refusal barely moves belief.        | Confidence-flat / mis-ordered ingestion.          |
| G5                                              | Concept lifts member items (AUC > 0.8) above its popularity projection; channel $\geq$ member-bag.          | Non-specific / popularity-parasite concepts.      |
| G6                                              | Wrong-user, shuffled, placebo, and duplicated answers gain $\approx$ nothing.                               | Cardinality confound; existential leakage.        |
| G8                                              | Replacing $\Sigma$ with $c\mathbf{I}$ degrades the deciding number; order- and re-ask-invariant.            | Decorative $\Sigma$ ; order-dependence.           |
| G9                                              | On a fixed user-independent question set the curve still rises from answer content.                         | Taste-leak: selection masquerading as ingestion.  |
| <i>One-time certifications (per instrument)</i> |                                                                                                             |                                                   |
| C1                                              | Implementations snap to published numbers on the canonical split (met; Multi-DAE failed and was dropped).   | Uncertified ruler / metric mismatch.              |
| C2                                              | G2 gain survives a held-out behavioural answer model + noise; geometry-answer firewall.                     | Circular answer model (self-simulator inversion). |
| C3                                              | $\geq 1$ study: rendered questions $\rightarrow$ graded human answers $\rightarrow$ NDCG (deferred).        | Simulator-only external-validity gap.             |

Table 1: **The instrument acceptance battery.** Per-build gates G0–G9 (there is no standalone G7; the answer-model-transfer check is folded into C2) run on every build; certifications C1–C3 are one-time per instrument. The three headline outcome gates are G0/G1/G2; the semantic and artifact-control gates G3–G9 and the transfer certification C2 close the false positives that pass the outcome gates alone (text). The full specification, with each gate’s testable form, is the design-side battery document.

| Family                               | Representatives                                      | R1       | R2       | R3       | R4       | R5       | One-line verdict                                                             |
|--------------------------------------|------------------------------------------------------|----------|----------|----------|----------|----------|------------------------------------------------------------------------------|
| (i) retrain / meta-learn per user    | MeLU, TaNP [20], CDRNP                               | ~        | ✓        | ✗        | ~        | ✗        | R2/R4 precedent; not full-profile SOTA, not channel-open.                    |
| (ii) fold-in linear / closed-form    | EASE [33], SLIM, GF-CF [30], BSPM [7], Turbo-CF [26] | ✓        | ✓        | ✗        | ✗        | ~        | The full-profile bar (G0); item-indicator basis $\Rightarrow$ R3 impossible. |
| (iii) amortised inference encoders   | Multi-VAE [19], RecVAE [31], EDLAE [34], EDDI [24]   | ✓        | ✓        | ~        | ~        | ✗        | The architecture we live in; VAEs item-only, EDDI right shape/wrong domain.  |
| (iv) Bayesian belief over item space | Bıyık [4], ConTS [17], BCIE [36], ICF [37]           | ✗        | ✓        | ~        | ✓        | ~        | The R4/R5 heart; belief layers, not full-profile towers.                     |
| (v) sequence models                  | SASRec [15], BERT4Rec [35]                           | ~        | ✓        | ✗        | ✗        | ✗        | Interview $\neq$ ordered session; tokens item-only, order-sensitive.         |
| (vi) LLM-as-recommender / CRS        | PEBOL [3], GATE [16], LLM-CRS [13]                   | ✗        | ✓        | ✓        | ~        | ✗        | Best R3 (open text); weakest R1 (weak collaborative signal).                 |
| <b>CASPER (target)</b>               | <b>this chapter</b>                                  | <b>✓</b> | <b>✓</b> | <b>✓</b> | <b>✓</b> | <b>✓</b> | <b>The conjunction; no published system occupies it.</b>                     |

Table 2: Six-family taxonomy scored against R1–R5. ✓ met, ~ partial, ✗ not met. Scores are our reading of each family’s *published* claims and are qualitative; the last row is a target, not a result. R1 and {R3,R4,R5} are anti-correlated: families (ii)/(iii) hold the accuracy corner, family (iv) the belief corner, and none spans both. Adapted from the landscape analysis (c2).



|  |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |    |
|--|----------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|----|
|  | measured |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | </ |
|--|----------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|----|

Table 3: **Master table: every reasonable system by R1–R5 and by measured full-profile accuracy on one shared ruler.**  $\checkmark$  met,  $\times$  not met (our reading of each system’s *published* claims, consistent with Table 2). *full/tail NDCG@10* is on the canonical Liang-recipe strong-generalization ML-25M split (catalog restricted to the 18,430-item project catalog, train vocabulary 18,359; train 140,768 / val 10k / test 10k users; per-user 80/20 fold-in; tail = top-33%-train-mass head mask, 190 head items), scored with the decoder’s learned bias. The Blyk and ConTS rows share one measured posterior core (belief-MF, best-effort; substitutions per the no-dash paragraph), so both print 0.2019/0.1200; the Golbandi row prints its node-recommender core. Statuses are *measured* (run on this ruler), *queued* (**TODO**: ], run pending), *queued* (**TODO**: ] : runnable, implementation/run scheduled on this ruler; best-effort = *superseded* (a measured successor row subsumes it), or *not runnable* / *not portable* (impossible by construction; mechanism in the text). The final column is each system’s own published full-profile anchor on its own split/metric—not comparable to our ruler until bridged (§7.2); “n/r” where the literature reports none. The RecVAE row is our own paper-config training run on this ruler and is the current full-profile frontier: the “ours” block adds the graded native interview tower T9/ whose certified run ties that frontier.

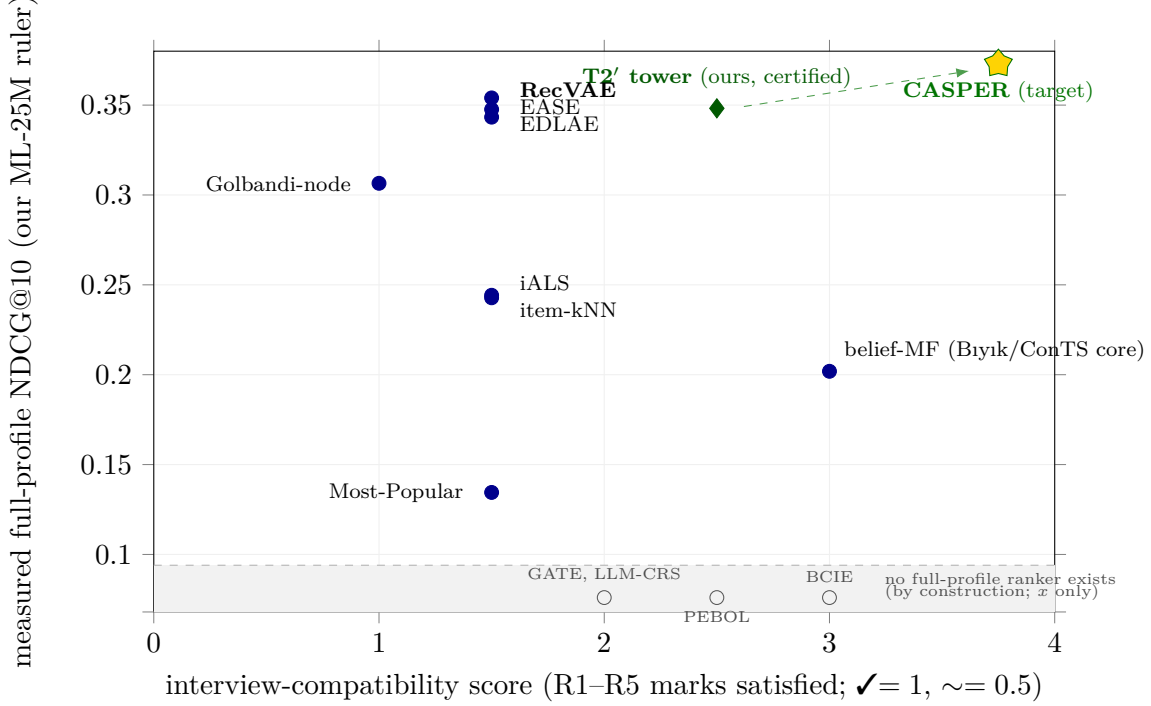


Figure 1: **The gap map (measured axes).** *y*-axis: measured full-profile NDCG@10 on our canonical ML-25M ruler. *x*-axis: an *interview-compatibility score*, defined as the number of **R2–R5** marks a system carries in the master table (Table 3), scoring ✓=1 and ~ = 0.5, maximum 4. *R1 is deliberately excluded from the score*: accuracy is the *y*-axis, and letting it also inflate *x* would blur the very trade-off the figure exists to show. The score is a *coarse ordinal capability count*, *not a measurement*, so the *x*-axis is qualitative even though the *y*-axis is measured. Filled markers are systems with a full-profile number on this ruler (belief-MF is the shared Biyık/ConTS posterior core; Golbandi-node is the tree’s best-effort node recommender). Hollow markers in the separated grey strip *below* the measured *y*-scale are systems that *cannot produce a full-profile number by construction* (no full-catalogue ranker, or not portable; see the master-table dispositions); only their *x* position is meaningful. Runnable-but-not-yet-measured systems (sequence models, graph filters, EDDI, TaNP) are deliberately *absent* from the figure — they carry **[TODO: ]** cells in Table 3 and join the plot when their numbers land. The star marks the instrument target (all five requirements at the frontier bar), still **[TODO: ]**; the green diamond is our *certified* tower T2’ at 0.3482—the first measured point that is simultaneously at the frontier bar and away from the item-only column, with the dashed arrow marking the remaining distance to the target (the channel and belief marks it has yet to earn in full). The population is otherwise *bimodal*: the entire accuracy corner (Pop through RecVAE, 0.343–0.354 at the top) collapses onto  $x \leq 1.5$ —strong recommenders that can fold items and nothing else—while every system with genuine interview machinery sits at  $x \geq 2$  and either has no measurable full-profile accuracy at all (the grey strip) or, for the one measurable core, drops to 0.2019, below plain matrix factorisation (belief-MF at  $x=3$ ). Two clusters, no bridge: the corner is not expensive; it is unoccupied. The nearest published rival to the target is Biyık et al. [4] (highest compatibility of any system with an actionable belief), whose runnable core nonetheless measures far below the frontier bar.

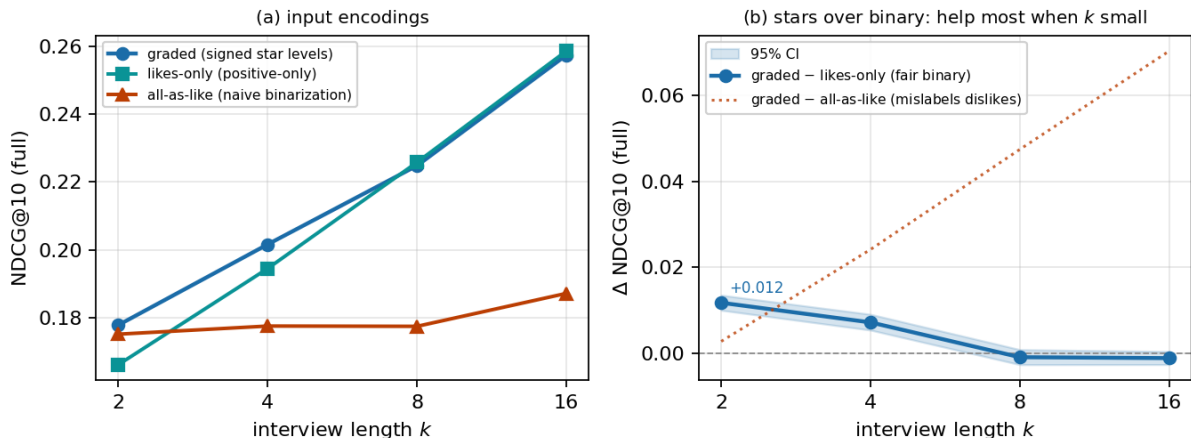


Figure 2: **Graded input helps over binarized input, most on the short interview (provisional; frozen graded-native tower, ML-25M interview harness, VAL, full NDCG@10 vs. interview length  $k$ ).** The same revealed set is fed three ways: *graded* (real signed star levels), *likes-only* (sub-threshold answers dropped—exactly what a positive-only tower such as RecVAE does at  $r>3.5$ ), and *all-as-like* (every answered item recorded as a 4-star like—naive implicit binarization). (b) Against the fair positive-only baseline, the graded premium is *CI-clean positive on the short interview*—+0.012 at  $k=2$  (CI [+0.010, +0.014]), +0.007 at  $k=4$ —and fades to a tie by  $k=8$  as collaborative signal fills in: the extra bits a star carries are most decisive when few items are known. The *mechanism* is that a signed answer encodes a dislike (a negative  $y_j$  the item-indicator basis cannot express); against the *all-as-like* control, which mislabels dislikes as likes, the apparent premium instead *grows* with  $k$  (dotted)—a control artifact (more revealed items  $\Rightarrow$  more dislikes corrupted), not value from gradation. Mirroring the star signs (flip) costs up to  $-0.29$ . Measured on the pre-certification snapshot of the tower architecture; **[RE-RUN QUEUED: graded-vs-binarized curve on the certified tower]**

| Baseline                                                     | Type                             | Reported anchor                            | Why the slot                                            |
|--------------------------------------------------------------|----------------------------------|--------------------------------------------|---------------------------------------------------------|
| <i>Tier 1 — classic floors (must clear)</i>                  |                                  |                                            |                                                         |
| Most-Popular                                                 | non-personalised                 | popb floor (our harness)                   | the “did we learn anything” line; G2 must beat it cold. |
| Item-kNN (cosine)                                            | neighbourhood                    | ML-20M NDCG@100<br>~ 0.36 class            | instant fold-in; Cremonesi-era strong-simple [8].       |
| PureSVD / iALS-WMF                                           | linear MF fold-in                | WMF ML-20M NDCG@100 0.386                  | cheap fold-in reference.                                |
| SLIM                                                         | sparse linear item-item          | ML-20M NDCG@100 0.401                      | direct EASE ancestor.                                   |
| <i>Tier 2 — full-profile SOTA (the R1 / G0 bar)</i>          |                                  |                                            |                                                         |
| EASE [33]                                                    | closed-form item-item            | NDCG@100 0.420                             | the bar to tie; item-only shows the R3 gap.             |
| RecVAE [31]                                                  | amortised VAE                    | NDCG@100 0.442                             | top of the ML-20M split; our in-house frontier.         |
| Mult-VAE [19]                                                | amortised VAE                    | NDCG@100 (ours 0.422) 0.426                | canonical amortised anchor; near-snap.                  |
| <i>Mult-DAE</i> (dropped)                                    | DAE                              | NDCG@100 (ours 0.397) 0.419                | snap <b>failed</b> -0.022; disqualified.                |
| EDLAE [34]                                                   | denoising higher-order EASE      | ≈ RecVAE class                             | confirms the flat frontier.                             |
| GF-CF / BSPM / Turbo-CF [30, 7, 26]                          | closed-form graph filter         | SOTA on Gowalla/Yelp (ML-20M not reported) | admit only if re-run on our split.                      |
| SASRec / BERT4Rec [15, 35]                                   | self-attentive sequence          | strong next-item (not this split)          | sequence fold-in; watch replicability [28].             |
| <i>Tier 3 — elicitation-compatible rivals (decide G1/G2)</i> |                                  |                                            |                                                         |
| EDDI / Partial-VAE [24]                                      | set-encoder + info-gain          | non-recsys (UCI/MNIST)                     | the R2 architecture; myopic info-gain baseline.         |
| Bıyık attributes [4]                                         | soft-Gaussian belief + EVOI      | own elicitation task                       | the central threat; frozen-vs-co-trained wedge.         |
| ConTS [17]                                                   | Thompson bandit, item+attr arms  | cold-start CRS metrics                     | unified-space + uncertainty; fixed schema.              |
| BCIE [36]                                                    | conjugate-Gaussian critique fold | critique metrics                           | exact-conjugacy-over-frozen precedent.                  |
| PEBOL [3]                                                    | Bayesian-opt NL-PE (LLM)         | MRR@10 0.27 vs 0.17                        | load-bearing PE baseline; per-item Beta.                |
| Golbandi tree [11]                                           | RMSE decision-tree interview     | Netflix RMSE                               | the static-tree interview comparator.                   |
| RBMF / Functional-MF [21, 38]                                | MF-embedding seeds               | elicitation RMSE                           | interview-seed line; cite-and-differentiate.            |

Table 4: Baseline bank. Anchors are on each system’s own split/metric and are *not* comparable to ours until the bridge (§7.2) is run; G0 is decided against EASE+RecVAE, G1/G2 against the Tier-3 rivals plus a random-question control. Any deep-net “SOTA” not snapping to Tier-2 numbers on our split is inadmissible [10].

| Method                              | full NDCG@10  | tail NDCG@10  | note                                            |
|-------------------------------------|---------------|---------------|-------------------------------------------------|
| Most-Popular                        | 0.1345        | 0.0226        | @100 0.1975; non-personalised floor             |
| belief-MF (Bıyık/ConTS core)        | 0.2019        | 0.1200        | @100 0.2886; best-effort posterior core         |
| Item-kNN (cosine)                   | 0.2428        | 0.1297        | @100 0.3200; neighbourhood fold-in              |
| iALS-WMF                            | 0.2442        | 0.1853        | @100 0.3704; MF fold-in                         |
| Golbandi node recommender           | 0.3065        | 0.1895        | @100 0.3958; best-effort tree core              |
| EDLAE [34]                          | 0.3433        | 0.2395        | @100 0.4330; val $p=0.4$                        |
| EASE [33]                           | 0.3476        | 0.2441        | @100 0.4375                                     |
| RecVAE (paper config; our run) [31] | <b>0.3540</b> | <b>0.2497</b> | @100 0.4537; nominal frontier                   |
| <i>Ours</i>                         |               |               |                                                 |
| Set-encoder tower T2' (certified)   | 0.3482        | 0.2462        | @100 0.4486; <b>G0 tie</b> (−0.0057, CI 0.0069) |

Table 5: G0 — full-profile, items-only, on the canonical Liang-recipe strong-generalization ML-25M split (catalog restricted to the 18,430-item project catalog, train vocabulary 18,359; train 140,768 / val 10k / test 10k users; per-user 80/20 fold-in; seed 98765), NDCG@10 (full and tail); tail = top-33%-train-mass head mask (190 head items); scored with the decoder’s learned bias. The belief-MF and Golbandi rows are best-effort cores of the belief-corner systems (§3). All rows reuse the canonical harness verbatim, so they are on one ruler.